

Roll No.

TEC-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
BASIC ELECTRONICS ENGINEERING**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) $(4362.25)_7 = (?)_2 = (?)_8$

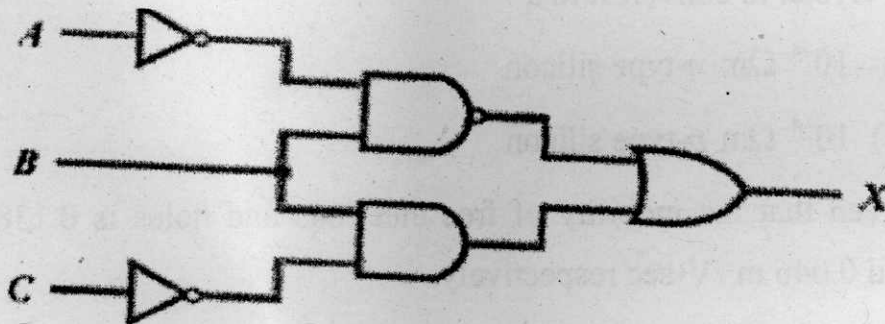
(ii) The solution of the quadratic equation $x^2 - 11x + 22 = 0$ are $x = 3$ or $x = 6$. What is the base of the numbers ? (CO1)

OR

(b) Minimize the following by K map and realize the minimized function by NAND gate only and NOR gate only : (CO1)

$$F(A, B, C, D) = \sum m(0, 2, 5, 8, 9, 10, 13) + \sum d(1, 7, 11)$$

2. (a) Derive Boolean expression for output X for given digital circuit and sketch truth table : (CO1)



P. T. O.

OR

- (b) Express $F(A, B, C, D) = (AB'X) + (A'B') + (ABC'D)$ in canonical SOP and canonical POS form and minimize the function by K-map.

(CO1)

3. (a) Derive an expression for drift and diffusion current density in semiconductor.

(CO2)

OR

- (b) A sample of Ge has intrinsic charge carrier density of $2.5 \times 10^{19}/\text{m}^3$. If it is doped with donor impurity of 1 atom in every 10^6 Ge atom. If the Ge atom density is $4.2 \times 10^{28}/\text{m}^3$, then determine the majority and minority charge carriers as well as conductivity and resistivity of the doped semiconductor.

(CO2)

Given that the mobility of free electrons and holes is $0.38 \text{ m}^2/\text{V-sec}$ and $0.18 \text{ m}^2/\text{V-sec}$ respectively.

4. (a) What is mobility and conductivity? Derive the equations for Mobility, Drift velocity, Resistivity, and Conductivity.

(CO2)

OR

- (b) Calculate the density of impurity atom that must be added to an intrinsic Si crystal to convert it to a :

(CO2)

(i) $10^{-4} \Omega\text{m}$ *n*-type silicon

(ii) $10^{-4} \Omega\text{m}$ *p*-type silicon

Given that the mobility of free electrons and holes is $0.138 \text{ m}^2/\text{V-sec}$ and $0.046 \text{ m}^2/\text{V-sec}$ respectively.

(3)

5. (a) Calculate the drift current produced in an intrinsic Ge semiconductor having area of cross section 1 cm^2 and length 0.3 mm , if concentration of free electron is $2 \times 10^{19}/\text{m}^3$ a battery of 2 volts is applied across its length. Given that mobility of free electrons and holes is $3500 \text{ cm}^2/\text{V-sec}$ and $1700 \text{ cm}^2/\text{V-sec}$ respectively. (CO1/CO2)

OR

- (b) State and prove De Morgan's theorem. Explain min-terms and max-terms. (CO1/CO2)

Roll No.

BTE-201

B. TECH. (BIOTECH.) (SECOND SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ADVANCED ORGANIC CHEMISTRY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the chlorination reaction of benzene with mechanism. (CO2)

OR

- (b) Elaborate the nitration of benzene with mechanism. (CO2)

2. (a) Explain the S_N^2 with mechanism explain which alkyl halide give S_N^2 reaction faster. (CO2)

OR

- (b) Evaluate the S_N^1 reaction with mechanism and explain which alkyl halide give S_N^1 reaction faster. (CO2)

3. (a) Define the following terms with suitable example : (CO2)

(i) Resonance

(ii) Hyperconjugation

OR

- (b) Explain the term Carbocation in terms of hybridization and stability.

(CO2)

P. T. O.

(2)

4. (a) How can you purify the given organic compound by crystallization method ? (CO1)

OR

- (b) Define the term sublimation with suitable example. (CO1)

5. (a) What do you mean by the term Chromatography ? Give the classification of chromatography. (CO1)

OR

- (b) Explain distillation method with suitable diagram. (CO1)

Roll No.

TCE-201

B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

BASIC CIVIL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Give definition of the term "Civil Engineering". Enlist different branches of Civil Engineering and discuss minimum two branches dealing with Civil Engineering design and construction. (CO1)

OR

- (b) We all know that the infrastructure plays an important role in the economic development of a nation, so can you list down 10 points to prove the above statement. (CO1)

2. (a) Write down Top 5 Civil Engineering marvels of India/World and provide details in the following format : (CO1)

Name	Location (Country)	Year of completion	Purpose	Important Features

P. T. O.

OR

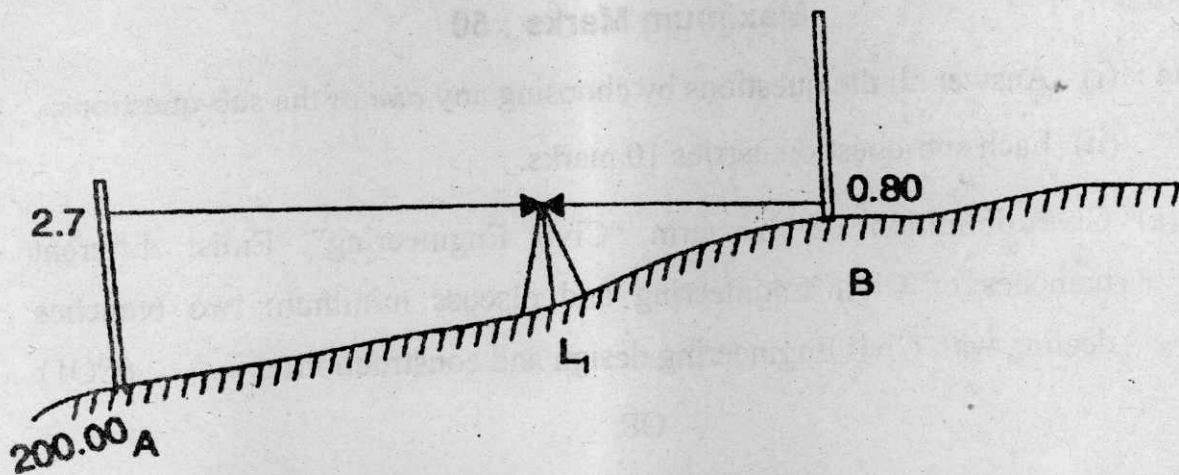
(b) Briefly discuss the following structures :

(CO1)

- (i) Bridge/Tower
- (ii) Retaining walls
- (iii) Dams
- (iv) Railways
- (v) Tunnel

3. (a) Find out the R. L. of Point B, with reference to the following figure :

(CO1)



OR

(b) Distinguish between the following :

(CO1)

- (i) Plane surveying and Geodetic surveying
- (ii) Quadrantal bearing and Whole circle bearing

4. (a) What is purpose of ranging in linear measurement and how can it be done ?

(CO2)

(3)

OR

- (b) In traversing in anticlockwise direction, the following readings were observed : (CO2)

Line	F. B.
AB	105° 15'
BC	20° 00'
CD	316° 30'
DE	187° 15'
EA	122° 45'

Draw a neat sketch of the traverse. Determine the interior angles of the traverse and apply check.

5. (a) Explain the working of Global Positioning System with the help of neat diagram. (CO2)

OR

- (b) Write a short note on "Various Area of Application of Remote Sensing" and how it is integrated with GIS. (CO2)

Roll No.

TCH-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

ENGINEERING CHEMISTRY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) (i) Differentiate between intramolecular and intermolecular hydrogen bonding.
(ii) Draw completely labelled MOT diagram of O_2^+ and N_2^+ and explain their magnetic property. (CO1)

OR

- (b) (i) Explain different types of vibrations involved in IR spectroscopy.
(ii) Enumerate the properties of nanoscale materials. (CO1)
2. (a) Explain Band Theory taking the example of Na metal. Comment on the metallic properties of Na. (CO1)

OR

- (b) (i) Differentiate between intrinsic and extrinsic semiconductors.
(ii) Explain the basic principle of UV-Vis spectroscopy. (CO1)

P. T. O.

(2)

3. (a) Describe the principle of ion exchange process of boiler water treatment with appropriate diagram and chemical equations.

Outline 2 advantages of the process vis-à-vis lime soda process of water treatment. (CO2)

OR

- (b) Calculate the amount of lime and soda required for the treatment of 100,000 litres of water containing the following impurities in ppm :
 $\text{Mg}(\text{HCO}_3)_2 = 20.8$, $\text{CaCl}_2 = 28.1$, $\text{MgCl}_2 = 8.78$, $\text{Ca}(\text{HCO}_3)_2 = 30.2$,
 $\text{CaSO}_4 = 35.0$, $\text{MgSO}_4 = 6.7$.

The purity of lime and soda used are 70% and 85% respectively. (CO2)

4. (a) Differentiate between the following : (CO2)
- (i) Cold lime soda and Hot lime soda
 - (ii) Temporary hardness and Permanent hardness

OR

- (b) A completely exhausted zeolite softner requires 120 litres of NaCl solution having 100 g/l of NaCl. How many litres of water having hardness 500 ppm can be softened by the zeolite softner ? (CO2)

5. (a) Differentiate between GCV and LCV of a fuel. How are they computed with the help of a bomb calorimeter ? (CO4)

OR

- (b) Calculate the net calorific value of the fuel if its ultimate analysis comes out to be containing C = 64%, O = 14.56%, H = 10.44%, N = 6.52%, S = 4.48%, Weight of coal burnt is 0.994 g; Water equivalent of calorimeter 3.94 kg, Rise in temperature = 2.7°C ; latent heat of condensation of steam 587 cal/g. (CO4)

Roll No.

TPH-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

ENGINEERING PHYSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) In Fresnel Biprism experiment, deduce the expression to determine the distance between two virtual sources using displacement method. (CO1)

OR

- (b) Derive the expression for path difference in a wedge shape film in reflected light also write the condition for maxima and minima. Explain why an extended source of light is required in thin films ? (CO1)

2. (a) In Newton's ring experiment derive the expression for diameter of bright rings and show that the spacing between the rings decreases as the order increases. (CO1)

OR

- (b) In Newton's ring arrangement a source emitting two wavelengths $\lambda_1 = 6000 \text{ \AA}$ and $\lambda_2 = 4500 \text{ \AA}$ is used. It is found that n th dark ring due to λ_1 coincides with $(n + 1)$ th dark ring due to λ_2 . If the radius of curvature of the plano convex lens is 100 cm, find the diameter of the n th ring for λ_1 . (CO1)

P. T. O.

(2)

3. (a) In N slits Fraunhofer diffraction, derive the expression for resultant intensity write the ratio for intensities of principal maxima and secondary maxima. (CO1)

OR

- (b) Derive the expression for resolving power of diffraction grating. What is the highest order spectrum seen with light of wavelength $6000 \text{ \AA}$ by means of grating having 5000 lines/cm . (CO1)

4. (a) Derive the expression for numerical aperture and acceptance angle in an optical fiber. (CO2)

OR

- (b) An optical fiber the refractive index of the core is 1.4 and the fractional refractive index change is 0.03. Find the refractive index of cladding and the numerical aperture. (CO2)

5. (a) Discuss the working principle of fiber optics also distinguish between various types of optical fiber. (CO2)

OR

- (b) In an optical fiber the refractive index of core and cladding are 1.55 and 1.50 respectively. Calculate the critical angle, acceptance angle and numerical aperture. (CO2)

Roll No.

BTS-201

B. TECH. (BIOTECHNOLOGY) (SECOND SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ENTREPRENEURIAL BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the different biosafety levels ? Elaborate in detail. (CO1)

OR

(b) What are the ethical challenges associated with transgenic plants ?
Discuss them in detail. (CO1)

2. (a) Elaborate on the concerns related to bio-safety with reference to
research in biotechnology. (CO1)

OR

(b) Discuss some of the ethical concerns related to Gene Therapy. (CO1)

3. (a) Elaborate on the environmental issues related to bioethics.

(CO1/CO2)

OR

(b) Elaborate on the mandate and tasks of the International Bioethics
Committee. (CO1/CO2)

P. T. O.

(2)

4. (a) Discuss some of the important bio-business in India. (CO2)

OR

- (b) Discuss some of the ethical concerns related to bio-business. (CO2)

5. (a) What is SWOT analysis ? Elaborate in detail .? (CO2)

OR

- (b) Explain Entrepreneurship. What are the different types of Entrepreneurship ? (CO2)

Roll No.

TEE-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, 2023**

(All Branches)

BASIC ELECTRICAL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Write statement of the following d.c. theorems with suitable example : (CO1)

(i) Thevenin's Theorem

(ii) Norton's Theorem

(iii) Maximum Power Transfer Theorem

OR

(b) Write the statement of KCL.

Find the resistance R_{AB} in Fig. 1 by using star-delta or delta-star transformation. (CO1)

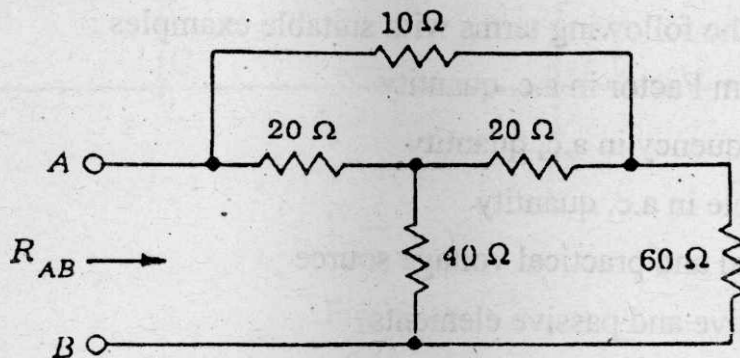


Fig. 1

P. T. O.

2. (a) Find the voltage between A and B shown in Fig. 2.

(CO1)

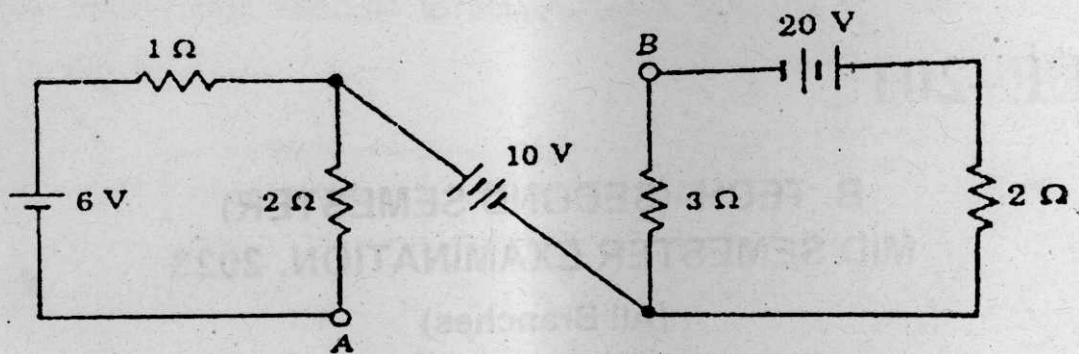


Fig. 2

OR

- (b) Determine the current through 5Ω resistor shown in Fig. 3 by using Thevenin's theorem.

(CO1)

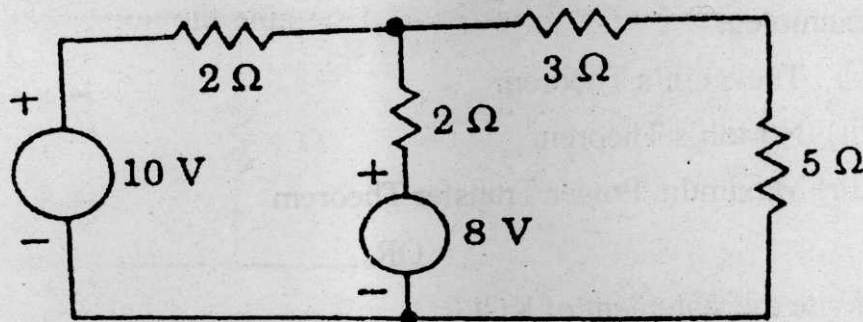


Fig. 3

3. (a) Define the following terms with suitable examples :

(CO1 & CO2)

- (i) Form Factor in a.c. quantity
- (ii) Frequency in a.c. quantity
- (iii) Cycle in a.c. quantity
- (iv) Ideal and practical voltage source
- (v) Active and passive elements

(3)

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OR

- (b) By using superposition theorem, determine current I across 3Ω resistor as shown in fig. 4. (CO1 & CO2)

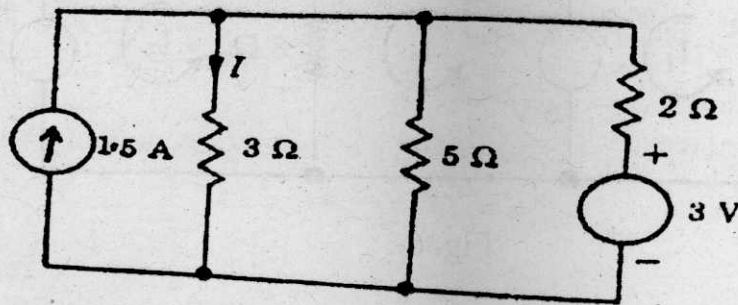


Fig. 4

4. (a) By using node-voltage analysis, determine the node voltages at node 1 and node 2 as shown in fig. 5. (CO1)

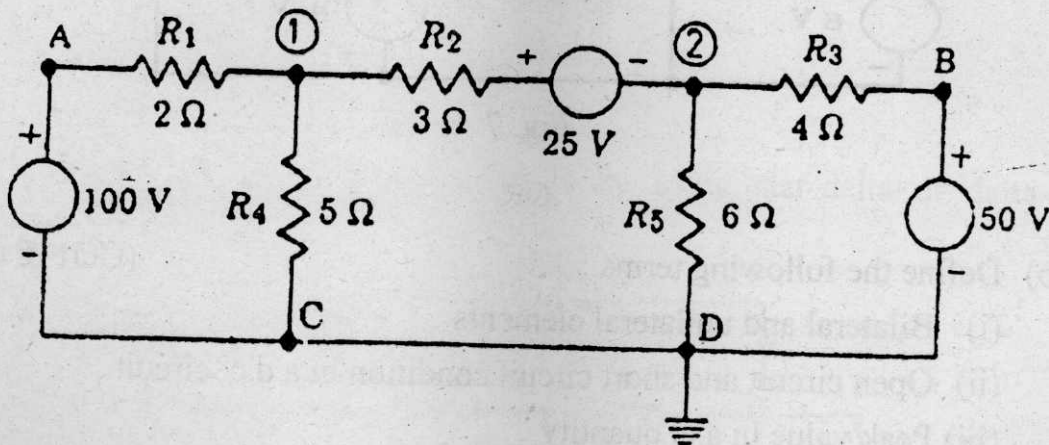


Fig. 5

P. T. O.

OR

- (b) Apply mesh analysis to determine the mesh currents as shown in Fig. 6. (CO1)

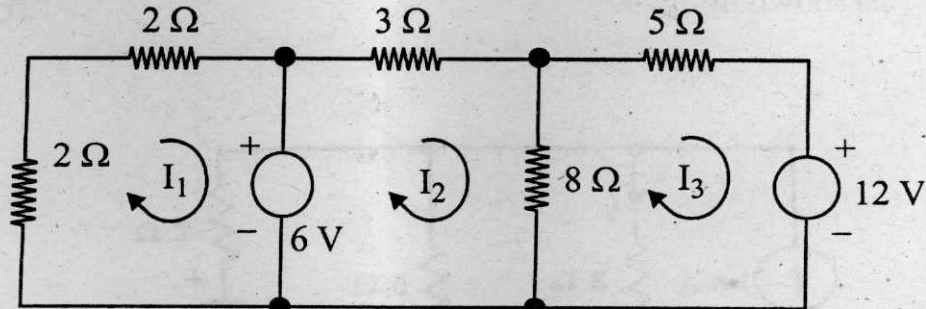


Fig. 6

5. (a) Write the statement of KVL with a suitable example. Use source transformation to determine the current across 1Ω resistor as shown in Fig. 7. (CO1 & CO2)

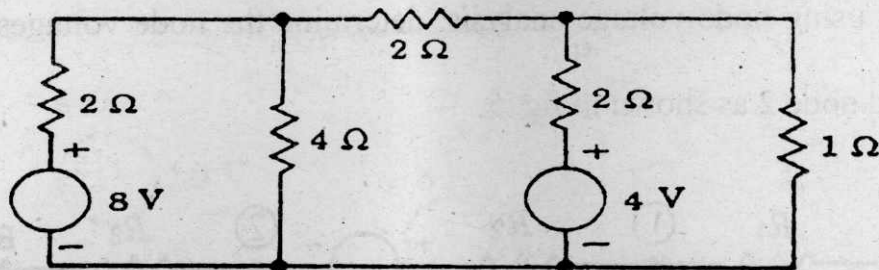


Fig. 7

OR

- (b) Define the following terms : (CO1 & CO2)
- Bilateral and unilateral elements
 - Open circuit and short circuit condition in a d.c. circuit
 - Peak value in a.c. quantity
 - Peak factor in a.c. quantity
 - Supermesh condition.

Roll No.

TCS-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

PROGRAMMING FOR PROBLEM SOLVING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

Attempt all parts of Question No. 2.

(ii) Each sub-question carries 10 marks.

1. (a) What is a string ? How is it different from character array ? What are the various methods to read a string ? Input a string and count number of spaces in that string. (CO1)

OR

- (b) Explain four library functions of strings with suitable syntax. Input a string, reverse it and print the final string. (CO1)

2. Find the output with explanation : (CO1 and CO3)

(i) #include <stdio.h>

int main()

{

char *p1 = "GOAT";

char *p2;

p2 = p1;

printf("%s", p2);

}

P. T. O.

(ii) #include <stdio.h>

int main()

{

char ch1[] = "hello";

char ch2[] = "hello";

if(ch1 == ch2)

printf("EQUAL");

else

printf("NOT EQUAL");

}

(iii) #include <stdio.h>

void cat(int *);

int main()

{

int i = 97, *p = &i;

cat(&i);

printf("%d", *p);

}

void cat(int *p)

{

int j = 2;

p = &j;

printf("%d", *p);

}

(iv) #include <stdio.h>

```
int main()
```

```
{
```

```
    float arr[5] = {12.5, 10.0, 13.5, 90.5, 0.5};
```

```
    float *ptr1 = &arr[0];
```

```
    float *ptr2 = ptr1 + 3;
```

```
    printf("%f", *ptr2);
```

```
    printf("%ld", ptr2 - ptr1);
```

```
    return 0;
```

```
}
```

(v) #include <stdio.h>

```
void geu(int*);
```

```
int main()
```

```
{
```

```
    int i = 10, *p = &i;
```

```
    geu(p++);
```

```
}
```

```
void geu(int *p)
```

```
{
```

```
    printf("%d\n", *p);
```

```
}
```

3. (a) Draw a flowchart to input a string and convert the lower-case alphabet to upper case alphabet and vice versa using pointer. (CO2)

OR

- (b) Draw a flowchart to input an array of n integer elements and insert an element k in a particular index with pointer. (CO2)

4. (a) What is a pointer ? Explain various types of pointer available in C. What do you mean by call by value and call by reference ? Explain with an example. (CO3)

OR

- (b) What do you mean by dynamic memory allocation ? What are the various functions to achieve this ? Write a C program to allocate memory for n integer elements dynamically and find sum of the elements of the array. (CO3)

5. (a) Write a program in C to pass two strings into a function using pointer and check whether the strings are anagram or not. (CO1 and CO3)

OR

- (b) Write a program in C to pass an array of n integer elements into a function using pointer and return the sum of all single digit and multi digit elements through a single function. (CO1 and CO3)

Roll No.

THU-201

B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
(All Branches)

ADVANCED PROFESSIONAL COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) There are 5 main questions and all must be attempted.
(ii) Each main question has 2 parts : (a) and (b) out of which only one part is to be attempted.
(iii) All questions are to be attempted in the given order only.

1. Write an essay of 300 words : (10 Marks) (CO1)

(a) A memorable journey

OR

(b) Is social media negatively impacting the youth ?

2. Write a paragraph of 150 words : (10 Marks) (CO1)

(a) If I could time travel I would...

OR

(b) Having hobbies are important for personality development.

P. T. O.

3. Write a Précis on the following : (10 Marks) (CO1)

- (a) Exercise is essential for maintaining physical health and can have many benefits for mental well-being as well. Regular physical activity can help reduce the risk of developing chronic conditions such as obesity, type 2 diabetes and heart disease. It can improve sleep, increase energy levels, and reduce stress and anxiety. Despite the many benefits of exercise, many people struggle to make it a regular part of their routine.

One reason for this could be that they don't know how to get started or don't have access to resources like a gym or personal trainer. However, there are many easy ways to incorporate exercise into daily life, such as walking or cycling, participating in team sports or trying home workout videos.

OR

- (b) Religion seems to me to be based primarily and mainly on fear. It is partly the terror of the unknown and the desire to feel that you have a kind of big brother who will stand by you in all your troubles and conflicts. Fear is the basis of the whole thing — fear of the mysterious, fear of failure, fear of death. Fear is the parent of cruelty and therefore, it is not surprising that cruelty and religion go together.

For fear in this world is the foundation of these two things, that we now understand certain things, and can master them with the help of science, which has forced its way step by step against Christianity: against the churches, and against all the opposition of the old doctrines.

Science can help us overcome this terrible fear that mankind has lived in for generations. Science can no longer teach us to look for allies in the sky but to look below at our own efforts to make this earth a habitable place.

4. Attempt as per the directions indicated below : (10 Marks) (CO2)

- (a) 1. John and George are *fair weather friends* as they always support each other in need.

(Correct the usage of italicized idiom)

2. The Police concluded that the evidence was *no more* with them.

(Provide a suitable oxymoron for the italicized phrase)

3. Suzanne was *raised and taken care of* by her uncle after her parents kicked the bucket.

(Provide a suitable phrasal verb for the italicized phrase)

4. He *worked hard* to clear the UGC NET.

(Replace the italicized phrase with an appropriate idiom)

5. The ensemble donned by Kate Winslet at Oscars was *terribly good*.

(Replace the italicized oxymoron with one meaning the opposite)

OR

- (b) 1. Stress has turned the today's youth into a _____.

(Fill in the blank using an appropriate oxymoron)

2. The actors were seen _____ with each other in the award function.

(Fill in the blank using an appropriate phrasal verb)

3. The doctors have _____ about the ethics of diagnosis and treatment.

(Fill in the blank with an idiom meaning started to understand)

P. T. O.

4. Singer KK did not _____ performing on the stage even during unfit medical condition.

(Fill in the blank using an appropriate oxymoron)

5. The release of the teaser was _____ because the censor board did not approve of it.

(Fill in the blank using an appropriate phrasal verb)

5. Attempt as per the directions against each part : (10 Marks) (CO2)

- (a) 1. Siddharth marrying Kiara was an _____ as fans were aware of it even before the pictures came out into the media officially.

(Fill in the blank using an appropriate oxymoron)

2. The professor was feeling _____ therefore he did not teach during the class.

(Fill in the blank using an appropriate idiom meaning ill)

3. The trip to Sweden turned into _____ when upon reaching there I received an email stating that I had to submit my final year project in merely a month's time.

(Fill in the blank using an appropriate oxymoron)

4. The man had to _____ his tears at his wife's funeral because of the presence of his daughter who loved her mother terribly.

(Fill in the blank using an appropriate phrasal verb)

5. The song that released last week was a _____ because the new single that got released yesterday is a super-hit.

(Fill in the blank using an appropriate idiom)

OR

- (b) 1. The sudden eruption of a dormant Volcano took lives of a lot of locals.

(Replace the italicized phrase with a suitable phrasal verb)

2. Harry and Hillary should now act mature and stop fighting the way they do.

(Replace the italicized phrase with an appropriate idiom)

3. Mr. Joe Biden was welcomed with a low level of noise by the US Senate. (Provide a suitable oxymoron for the italicized phrase)
4. If a student feels that he can manage passing in an examination without studying, he/she is sadly mistaken.

(Provide a suitable phrasal verb for the italicized word without altering the meaning of the sentence)

5. All the celebrities are confused as to how to managed stardom.

(Replace the italicized word with an appropriate idiom meaning the same)

Roll No.

MAB-201

B. TECH. (BIOTECHNOLOGY) (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
ENGINEERING MATHEMATICS

(Chemistry Group)

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define Matrix and all its types with counter examples. (CO1)

OR

(b) Define Determinant. What is the difference between Determinant and Matrices ? (CO1)

2. (a) Evaluate the following limits : (CO2)

(i) $\lim_{x \rightarrow \infty} \frac{9x^2 - x + 5}{8x^2 + 7x + 9}$

(ii) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

OR

(b) A function is defined as follows. Using limits, examine its continuity at $x = 1$ and find $f(1/2)$: (CO2)

$$\begin{aligned} f(x) &= 1 - x && \text{when } 0 \leq x \leq 1 \\ &= x - 1 && \text{when } 1 < x \end{aligned}$$

P. T. O.

(2)

3. (a) Show that the following matrix is its own adjoint : (CO1)

$$\begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$$

OR

- (b) Solve the following system of equations by matrix method : (CO1)

$$2x - y + 3z = 9$$

$$x + y + z = 6$$

$$x - y + z = 2$$

4. (a) What will be the values of x for the maximum and the minimum values of the following function $y = x^3 + 2x^2 - 4x - 8$? (CO2)

OR

- (b) Find $\frac{dy}{dx}$ where : (CO2)

(i) $y = \frac{x^2 + 5x}{\sqrt{x}}$

(ii) $y = e^x \log x$

5. (a) Evaluate the following : (CO2)

(i) $\int \frac{2x-2}{x^2-2x+3} dx$

(ii) $\int \log x dx$

OR

- (b) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$. (CO1)

Roll No.

MAB-201

B. TECH. (BIOTECHNOLOGY) (SECOND SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ENGINEERING MATHEMATICS

(Chemistry Group)

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define Matrix and all its types with counter examples. (CO1)

OR

- (b) Define Determinant. What is the difference between Determinant and Matrices ? (CO1)

2. (a) Evaluate the following limits : (CO2)

(i) $\lim_{x \rightarrow \infty} \frac{9x^2 - x + 5}{8x^2 + 7x + 9}$

(ii) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

OR

- (b) A function is defined as follows. Using limits, examine its continuity at $x = 1$ and find $f(1/2)$: (CO2)

$$\begin{aligned} f(x) &= 1 - x && \text{when } 0 \leq x \leq 1 \\ &= x - 1 && \text{when } 1 < x \end{aligned}$$

P. T. O.

(2)

3. (a) Show that the following matrix is its own adjoint : (CO1)

$$\begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$$

OR

- (b) Solve the following system of equations by matrix method : (CO1)

$$2x - y + 3z = 9$$

$$x + y + z = 6$$

$$x - y + z = 2$$

4. (a) What will be the values of x for the maximum and the minimum values of the following function $y = x^3 + 2x^2 - 4x - 8$? (CO2)

OR

- (b) Find $\frac{dy}{dx}$ where : (CO2)

(i) $y = \frac{x^2 + 5x}{\sqrt{x}}$

(ii) $y = e^x \log x$

5. (a) Evaluate the following : (CO2)

(i) $\int \frac{2x-2}{x^2-2x+3} dx$

(ii) $\int \log x dx$

OR

- (b) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$. (CO1)

Roll No.

TMA-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

ENGINEERING MATHEMATICS-II

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) All questions are compulsory.
(ii) Answer the questions by choosing any *one* of the sub-questions.
(iii) Each sub-question carries 10 marks.

1. (a) Solve : (CO1)

$$(D^2 - 4D + 4)y = 4e^{5x} + \cos 5x;$$

$$D \equiv \frac{d}{dx}.$$

OR

(b) Solve : (CO1)

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 4y = \cos(\log x) + \sin(\log x).$$

2. (a) Solve : (CO1)

$$(2x \log x - xy) dy + 2y dx = 0$$

P. T. O.

OR

(b) Solve :

(CO1)

$$(y^3 - 2yx^2) dx + (2xy^2 - x^3) dy = x$$

3. (a) Solve by method of variation of parameters :

(CO1 & CO2)

$$\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}.$$

OR

(b) Find the Laplace transform of function :

(CO1/CO2)

$$f(t) = \frac{(e^{at} - \cos bt)}{t}.$$

4. (a) Find the Laplace transform of :

(CO2)

$$f(t) = \begin{cases} \frac{t}{k}, & \text{when } 0 < t < k \\ 1, & \text{when } t > k \end{cases}$$

OR

(b) Find the Laplace transform of :

(CO2)

$$\int_0^{\infty} te^{-3t} \sin t \, dt$$

(3)

5. (a) Using Convolution theorem, find inverse Laplace transform of the function : (CO2)

$$f(s) = \frac{s}{(s^2 + 1)(s^2 + 4)}$$

OR

- (b) Using Laplace transform, solve the following initial value problem : (CO2)

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 2y = 5 \sin x$$

$$y(0) = 0, y'(0) = 0.$$